

Q2. Manage Docker containers and images

Manage Docker containers and images: list, start, stop, remove containers, tag and push images to a registry, and use Docker Compose for multi-container setups.

- Check Docker is working

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker --version
docker compose version
Docker version 29.7.2, build a7dcaa6
Docker Compose version v5.4.0
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 10 minutes	0.0.0.0:5000->5000/tcp, [::]:5000...	docker-demo-container

List containers

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 10 minutes	0.0.0.0:5000->5000/tcp, [::]:5000...	docker-demo-container

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 10 minutes	0.0.0.0:5000->5000/tcp, [::]:5000...	docker-demo-container

Start a container

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker run -d --name mycontainer nginx
70f85dd876823dab8ee28efe6828c7103f41f9e99bec580e22364f0b8e861c45
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
70f85dd87682	nginx	"/docker-entrypoint..."	7 seconds ago	Up 7 seconds	80/tcp	mycontainer
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 10 minutes	0.0.0.0:5000->5000/tcp, [::]:5000...	docker-demo-container

- Stop the container

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker stop mycontainer
mycontainer
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
70f85dd87682	nginx	"/docker-entrypoint..."	22 seconds ago	Exited (0) 8 seconds ago	
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 11 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp

- Start the stopped container

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker start mycontainer
mycontainer
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
70f85dd87682	nginx	"/docker-entrypoint..."	About a minute ago	Up 6 seconds	80/tcp
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 11 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp

- Remove the container

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker stop mycontainer
mycontainer
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker rm mycontainer
mycontainer
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 12 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-demo-container

- List Docker images

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U
nginx:latest	b34848eff6db	241MB	66.2MB	
python:3.12-slim	09f7da3bc104	179MB	45.4MB	

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo$ docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	U
nginx:latest	b34848eff6db	241MB	66.2MB	
python:3.12-slim	09f7da3bc104	179MB	45.4MB	

```
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker pull nginx
Using default log: latest
Latest: Pulling from library/nginx
f340c1b7c7d6: Pull complete
b26c93c66e62: Pull complete
a44b5c6be616: Pull complete
d3e7asb3e834: Pull complete
c13f794dea35: Pull complete
87db7b726498: Pull complete
76727c824218: Download complete
25282a7645eb: Download complete
Latest: Downloaded newer image for nginx latest
Status: Downloaded newer image for nginx latest
docker.io/library/nginx:latest
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
docker-demo:1.0	3667be2e461d	222MB	55.1MB		
hello-world:latest	5dd8d3e6c255	25.9KB	9.49KB		
nginx:latest	05b8cb60c354	253MB	69.2MB		
python:3.12-slim	0577da3bc354	179MB	69.4MB		

- Tag an image

```
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker tag nginx:latest ayushi2222/my-nginx:1.0
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
docker-demo:1.0	3667be2e461d	224MB	55.1MB		
hello-world:latest	5dd8d3e6c255	25.9KB	9.49KB		
nginx:latest	05b8cb60c354	253MB	69.2MB		
python:3.12-slim	0977da3bc10N	179MB	69.4MB		
ayushi2222/my-nginx:1.0	05b8cb60c354	253MB	69.2MB		

- Login to Docker Hub

```
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker login

USING WEB-BASED LOGIN

Info • To sign in with credentials on the command line, use "docker login -u <username>"

Your one-time device confirmation code is: MDVN-HNSF
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser.
```

- Push the image

```
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker push ayushi2222/my-nginx:1.0
1.0
The push refers to repository [docker.io/ayushi2222/my-nginx]
0320c61b6753: Unavailable
6320c61b6753: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
88e7a65e5583: Unavailable
Push access access, repository does not exist or any specific authorization: server message: insufficient_scope: authorization failed
```

- Verify the image

```
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
docker-demo:1.0	3667be2e461d	224MB	55.1MB		
hello-world:latest	5dd8d3e6c255	25.9KB	9.49KB		
nginx:latest	05b8cb60c354	223MB	69.2MB		
python:3.12-slim	09f7da3bc10M	179MB	45.4MB		
ayushi2222/my-nginx:1.0	05b8cb60c354	253MB	69.2MB		

- Create a Compose project

```
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
docker-demo:1.0	3667be2e461d	224MB	55.1MB		
hello-world:latest	5dd8d3e6c255	25.9KB	9.49KB		
nginx:latest	05b8cb60c354	223MB	69.2MB		
python:3.12-slim	09f7da3bc10M	179MB	45.4MB		
ayushi2222/my-nginx:1.0	05b8cb60c354	253MB	69.2MB		

```
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo$ mkdir compose-demo
cd compose-demo
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo/compose-demo$ nano docker-compose.yml
ayushi2222@LAPTOP-SLVIDNHT:~/docker-demo/compose-demo$
```

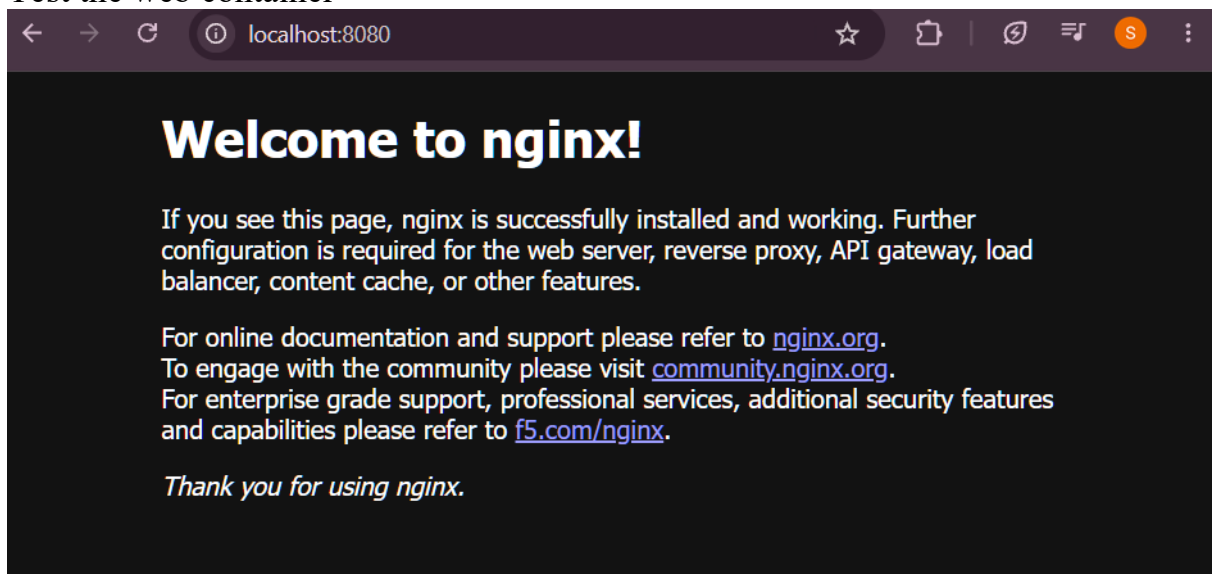
```
GNU nano 8.7.1 docker-compose.yml *
services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"
    depends_on:
      - redis

  redis:
    image: redis:latest
```

- Start both containers

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose up -d
[+] up 12/12
✓ Image redis:latest Pulled 9.6s
✓ Network compose-demo_default Created 0.2s
✓ Container compose-demo-redis-1 Started 1.5s
✓ Container compose-demo-web-1 Started 1.9s
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose ps
NAME                IMAGE             COMMAND             SERVICE    CREATED     STATUS      PORTS
compose-demo-redis-1 redis:latest      "docker-entrypoint.s_ redis      7 mins ago Up 6 seconds 6379/tcp
compose-demo-web-1  nginx:latest     "/docker-entrypoint. web        7 mins ago Up 5 seconds 0.0.0.0:8080->80/tcp, [:8080->80/tcp
```

- Test the web container



```

ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ curl http://localhost:8080
<!DOCTYPE html>
<title>Welcome to nginx!</title>
<p>If you see this page, nginx is successfully installed and working.<p>Further configuration is required for the web server, API gateway, load balancer.<a href="https://community.nginx.org/">community.service.org</a>.<br> For enterprise features and capabilities please refer to the documentation.<a href="https://trynginx.com/">trynginx.com</a>.</p>
ayushi2323@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose logs
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker processes
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 30
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 31
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 32
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 33
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 36
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 37
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 38
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 40
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:11 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36" "-"

```

- Check the Compose logs

```

web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/09/07 14:54:40 [notice] 1#1: using the "epoll" event method
web-1 | 2026/09/07 14:54:40 [notice] 1#1: nginx/1.31.5
web-1 | 2026/09/07 14:54:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/09/07 14:54:40 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
web-1 | 2026/09/07 14:54:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker processes
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 30
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 31
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 32
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 33
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 34
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 35
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 36
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 37
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 38
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 39
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 40
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 41
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:11 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36" "-"
redis-1 | Starting Redis Server
redis-1 | 1:C 07 Sep 2026 14:54:39.929 * o000o000o000o Redis is starting o000o000o000o
redis-1 | 1:C 07 Sep 2026 14:54:39.930 * Redis version=8.10.1, bits=64, commit=00000000, modified=1, pid=1, just started
redis-1 | 1:C 07 Sep 2026 14:54:39.930 * Configuration loaded
redis-1 | 1:M 07 Sep 2026 14:54:39.931 * monotonic clock: POSIX clock_gettime
redis-1 | 1:M 07 Sep 2026 14:54:39.937 * Running mode=standalone, port=6379.
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> RedisBloom version 8.10.1 (Git=unknown)
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> Registering configuration options: [
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-error-rate : 0.01 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-initial-size : 100 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-expansion-factor : 2 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-bucket-size : 2 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-initial-size : 1024 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-iterations : 20 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-expansion-factor : 1 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-expansions : 32 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> ]
redis-1 | 1:M 07 Sep 2026 14:54:39.939 * Module 'bf' loaded from /usr/local/lib/redis/modules/redisbloom.so
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> search-workers default: 12 (min of MAX_WORKER_THREADS=16 and CPU cores)
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> Redis version found by RedisSearch : 8.10.1 - oss
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> RedisSearch version 8.10.0 (Git=)
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read size: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results: 1000000, default scorer: BM25STD,
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Initialized thread pools!
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Enabled workers threadpool of size 12
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to config changes
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to cluster slot migration events
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to cluster topology change events
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Enabled role change notification
redis-1 | 1:M 07 Sep 2026 14:54:39.976 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
redis-1 | 1:M 07 Sep 2026 14:54:39.978 * Module 'search' loaded from /usr/local/lib/redis/modules/redisearch.so
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> RedisTimeSeries version 8.1000, git_sha=
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> Redis version found by RedisTimeSeries : 8.10.1 - oss
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> Registering configuration options: [
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-compaction-policy : }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-num-threads : 3 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-libmr-protocol : INTERNAL }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-retention-policy : 0 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-duplicate-policy : block }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-chunk-size-bytes : 4096 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-encoding : compressed }
redis-1 | 1:M 07 Sep 2026 14:54:39.980 * <timeseries> { ts-ignore-max-time-diff: 0 }

```

```

ayush12329@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose logs web
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/15-local-resolvers.eesh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/09/07 14:54:40 [notice] 101: using the "epoll" event method
web-1 | 2026/09/07 14:54:40 [notice] 101: nginx/1.21.5
web-1 | 2026/09/07 14:54:40 [notice] 101: built by gcc 14.2.0 (Debian 14.2.0-10)
web-1 | 2026/09/07 14:54:40 [notice] 101: OS: Linux 6.10.33.2-microsoft-standard-WSL2
web-1 | 2026/09/07 14:54:40 [notice] 101: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker processes
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 30
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 31
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 32
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 33
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 34
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 35
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 36
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 37
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 38
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 39
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 40
web-1 | 2026/09/07 14:54:40 [notice] 101: start worker process 41
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:11 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
Gacko) Chrome/152.0.0.0 Safari/537.36" "-"
web-1 | 2026/09/07 14:55:11 [error] 51831: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.18.0.1, server:
localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8888", referer: "http://localhost:8888/"
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:11 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8888/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gacko) Chrome/152.0.0.0 Safari/537.36" "-"
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:27 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.10.0" "-"
ayush12329@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose logs redis
redis-1 | Starting Redis Server
redis-1 | 1:C 07 Sep 2026 14:54:39.929 * oO0oO0oO0oO0o Redis is starting oO0oO0oO0oO0o
redis-1 | 1:C 07 Sep 2026 14:54:39.938 * Redis version 8.10.1, bits=64, commit=00080808, modified=1, pid=1, just started
redis-1 | 1:C 07 Sep 2026 14:54:39.938 * Configuration loaded
redis-1 | 1:M 07 Sep 2026 14:54:39.931 * monotonic clock: POSIX clock_gettime
redis-1 | 1:M 07 Sep 2026 14:54:39.937 * Running mode=standalone, port=6379
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> RedisBloom version 8.10.1 (Git:unknown)
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> Registering configuration options: [
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-error-rate : 0.01 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-initial-size : 100 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-expansion-factor : 3 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-bucket-size : 2 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-initial-size : 1024 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-iterations : 20 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-expansion-factor : 1 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-expansions : 32 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> ]
redis-1 | 1:M 07 Sep 2026 14:54:39.939 * Module 'bf' loaded from /usr/local/lib/redis/modules/redisbloom.so
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> search-workers default: 32 (min of MAX_WORKER_THREADS=16 and CPU cores)
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> Redis version found by RedisSearch: 8.10.1 - oss
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> RedisSearch version 8.10.0 (Git)
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> gc: 0M, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms):
500, simeous policy: return, oom policy: return, cursor read size: 1000, cursor max idle (ms): 300000, max dictable size: 1000000, max number of search res
ults: 1000000, default scorer: BM25TFIDF
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Initialised thread pool!

```

• Stop the Compose setup

```

srishti12902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose stop
[*] stop 2/2
✓Container compose-demo-web-1 Stopped 0.6s
✓Container compose-demo-redis-1 Stopped 0.3s
srishti12902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS

```

• Start the Compose setup again

```

srishti12902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose start
[*] start 2/2
✓Container compose-demo-redis-1 Started 0.5s
✓Container compose-demo-web-1 Started 0.5s
srishti12902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
compose-demo-redis-1 redis:latest "docker-entrypoint.s..." redis 2 minutes ago Up 5 seconds 6379/tcp
compose-demo-web-1 nginx:latest "docker-entrypoint..." web 2 minutes ago Up 5 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

```

• Remove the Compose containers

```

srishti12902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker compose down
[*] down 3/3
✓Container compose-demo-web-1 Removed 0.4s
✓Container compose-demo-redis-1 Removed 0.4s
✓Network compose-demo_default Removed 0.3s
srishti12902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
45a04fd4de37 docker-demo:1.0 "python app.py" 5 hours ago Up 21 minutes 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp docker-demo-container

```